

Online Supplement to

Dynamic Structural Equation Modeling in Blimp

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This supplement accompanies *Dynamic Structural Equation Modeling in Blimp*. Section S1 explains the factored-regression representation, and Section S2 presents a sensitivity analysis with person-specific linear trends. Complete data, executable Blimp and R scripts, and fitted output are provided in the Analysis-Example folder at <https://github.com/blimp-stats/dsem-tutorial>.

S1 DSEM Through Factored Regression

This section connects familiar DSEM equations to their factored regression representation. I begin with a basic autoregressive DSEM and then express its within-person and between-person components as conditional regressions, noting along the way why the person-specific mean is estimated as a latent quantity. For fuller treatments of factored regression, see Keller (2025) on latent variable interactions and nonlinear effects and Enders and Keller (2026) on factored structural equation modeling in Blimp. An introductory treatment of factored regression models is also available in Keller (2022).

S1.1 *A Basic Autoregressive DSEM*

Consider a single outcome Y_{tj} observed at time t for person j . A first-order autoregressive model represents the current outcome as a function of the person's typical level and their deviation at the prior occasion.

$$Y_{tj} = \beta_{0j} + \phi (Y_{t-1,j} - \beta_{0j}) + e_{tj} \quad (\text{S1})$$

In Equation (S1), β_{0j} is person j 's typical level, ϕ is the autoregressive coefficient, and e_{tj} is a within-person residual: the portion of the current outcome not predicted by the person's typical level and prior deviation. In time-series terminology, this residual is called an innovation. The predictor $Y_{t-1,j} - \beta_{0j}$ is the prior occasion's deviation from the person's typical level. Thus, ϕ describes how much of a prior deviation carries forward to the current occasion.

Subtracting the person mean from both sides makes the within-person process explicit.

$$Y_{tj} - \beta_{0j} = \phi (Y_{t-1,j} - \beta_{0j}) + e_{tj} \quad (\text{S2})$$

Equation (S2) shows that the current deviation from the person's typical level is predicted

by the prior deviation. At level 2, the person-specific means may vary around a grand mean.

$$\beta_{0j} = \mu_{\beta} + r_{\beta j} \quad (\text{S3})$$

In Equation (S3), μ_{β} is the grand mean and $r_{\beta j}$ is person j 's departure from it. Together, Equations (S1) and (S3) provide the familiar level-1 and level-2 pieces of a basic DSEM. The tutorial later fits this model to daily sleep quality and then expands it one component at a time.

In practice, β_{0j} is estimated rather than replaced by an observed person mean. In short series, treating the observed person mean as fixed can bias the estimated carry-over downward (Nickell, 1981; Hamaker & Grasman, 2015); estimating the person mean as a latent quantity avoids this observed-centering bias and propagates uncertainty about each person's typical level into the dynamic effects (Asparouhov, Hamaker, & Muthén, 2018; Asparouhov & Muthén, 2019). This approach is sometimes referred to as “latent centering” because the model constructs $Y_{t-1,j} - \beta_{0j}$ using the latent person mean rather than treating a precomputed deviation as fixed.

S1.2 *Expressing the DSEM as Conditional Regressions*

A key feature of the factored regression specification is that it avoids working with off-the-shelf multivariate distributions such as the multivariate normal distribution. Instead, the factored regression framework chains regression models (i.e., conditional densities) to characterize the hypothesized joint distribution. Throughout the tutorial, I represent models using a so-called “functional” notation. For the basic DSEM, the functional notation representation of the joint distribution of the current outcome and latent person mean, conditional on the prior outcome, is as follows.

$$f(Y_{tj}, \beta_{0j} | Y_{t-1,j}) = \underbrace{f(Y_{tj} | Y_{t-1,j}, \beta_{0j})}_{\text{within-person model}} \times \underbrace{f(\beta_{0j})}_{\text{between-person model}} \quad (\text{S4})$$

In Equation (S4), $f(\dots)$ represents a generic probability distribution for the variables inside the parentheses, and variables following the vertical bar are treated as conditioning variables. The left side represents the joint distribution of the current outcome and latent person mean, conditional on the prior outcome. Applying the chain rule factors this joint distribution into a within-person transition model and a between-person model for the person mean. The first density corresponds to Equation (S1), whereas the second corresponds to Equation (S3).

These two component densities map directly onto the within-person and between-person regression equations.

$$\begin{aligned} f(Y_{tj} | Y_{t-1,j}, \beta_{0j}) &\rightarrow Y_{tj} = \beta_{0j} + \phi(Y_{t-1,j} - \beta_{0j}) + e_{tj} \\ f(\beta_{0j}) &\rightarrow \beta_{0j} = \mu_{\beta} + r_{\beta j} \end{aligned} \tag{S5}$$

The first line of Equation (S5) specifies a linear autoregressive equation, and the second specifies an intercept-only between-person equation. These equations, rather than the functional factorization alone, imply the specific regression models used here. Retaining the factorization while changing the corresponding equation allows a component to be represented by a different regression model. For example, a conditional equation could involve transformed variables, could be specified as an ordinal regression, or—as shown later in the tutorial—could be expanded to include interaction terms among its predictors. The tutorial begins with the linear autoregressive specification and then changes the equations to accommodate multiple outcomes, person-specific coefficients, and person-specific residual variances while retaining the same factored-regression logic.

S2 Sensitivity to Person-Specific Linear Trends

The six tutorial models describe sleep and pain as fluctuating around stable person-specific trait levels. This assumption defines the lagged predictors as deviations from those levels. In applications where an outcome rises or falls systematically over the study, part of that trajectory can be mistaken for short-term carry-over. A sensitivity analysis can replace each stable level with a person-specific trajectory and define the lagged predictors relative to that fitted trajectory. This is a linear residual DSEM specification, in which short-term dynamics operate among deviations from person-specific trends (Asparouhov et al., 2018).

Let d_{Sj} and d_{Pj} denote person-specific linear trends in sleep and pain. The detrended lagged deviations subtract each person's fitted value on the prior day.

$$\begin{aligned}\text{sleep}_{t-1,j}^* &= \text{sleep}_{t-1,j} - [\beta_{0j} + d_{Sj}\text{day}_{t-1}] \\ \text{pain}_{t-1,j}^* &= \text{pain}_{t-1,j} - [\gamma_{0j} + d_{Pj}\text{day}_{t-1}]\end{aligned}\tag{S6}$$

The current sleep and pain equations include each person's trajectory on day t , whereas the lagged dynamic terms use the deviations on day $t - 1$. The four carry-over and spillover coefficients can remain random, and the fixed sleep-by-pain interaction can be formed from the two detrended lagged deviations. The names `sleep_resid_lag` and `pain_resid_lag` distinguish these trajectory residuals from the trait-centered lags in the primary models.

LATENT:

```
person = b_0j b_1j b_2j
        g_0j g_1j g_2j
        d_Sj d_Pj;
```

MODEL:

```
sleep_resid_lag = sleep.lag
                 - (b_0j + d_Sj*(day - 1));
pain_resid_lag = pain.lag
                - (g_0j + d_Pj*(day - 1));
```

```

within:
  sleep ~ intercept@b_0j day@d_Sj
         sleep_resid_lag@b_1j
         pain_resid_lag@b_2j
         (sleep_resid_lag*pain_resid_lag)@b_3;

  pain ~ intercept@g_0j day@d_Pj
         pain_resid_lag@g_1j
         sleep_resid_lag@g_2j;

```

The excerpt shows the latent declarations and within-person equations. The complete specification, including the between-person model and sampler settings, is in `Analysis-Example/detrended-sensitivity.imp`; its fitted output is in `Analysis-Example/detrended-sensitivity.blimp-out`.

Results and Interpretation

For the detrended model, the MCMC chains were well behaved during warm-up. In the final warm-up window, the largest rank-normalized $\hat{R}$ across parameters was 1.006, below the 1.05 criterion. The retained draws also provided stable posterior summaries; the smallest ESS among the five coefficients discussed below was 1,577. Accordingly, I focus on the two average linear trends, average sleep and pain carry-over, and the focal interaction. Complete results are available in the accompanying output file.

Beginning with the linear trends, sleep quality changed by an average of 0.003 points per day, with a 95% credible interval from -0.013 to 0.020 . The interval included zero, providing little evidence of a common direction of linear change in sleep quality. In contrast, pain increased by an average of 0.029 points per day, with a 95% credible interval from 0.012 to 0.048 . These coefficients describe average change across participants; the model allows each participant to have a different sleep and pain trajectory.

After accounting for those trajectories, average sleep carry-over was 0.134, with a 95% credible interval from 0.075 to 0.195 . When prior-day pain was at the person's fitted trajectory, a one-point higher prior-day sleep score relative to the person's sleep trajectory was

associated with a 0.134-point higher current sleep score, controlling for prior-day pain deviations. Average pain carry-over was 0.153, with a 95% credible interval from 0.082 to 0.225. Thus, a one-point higher prior-day pain score relative to the person's pain trajectory was associated with a 0.153-point higher current pain score, controlling for prior-day sleep deviations. Both outcomes showed positive average carry-over around the fitted trajectories.

The interaction coefficient was $\beta_3 = -0.039$, with a 95% credible interval from -0.063 to -0.014 . Controlling for the lower-order sleep carry-over and pain-to-sleep spillover associations, each one-point increase in prior-day pain relative to the person's fitted pain trajectory was associated with a 0.039 reduction in average sleep carry-over. Higher relative pain was therefore associated with weaker average sleep carry-over, consistent with the Model 3 estimate of -0.031 , with a 95% credible interval from -0.054 to -0.007 . The direction and magnitude of the interaction were similar after allowing for person-specific linear trends. The interpretation now concerns deviations from those trajectories, however, rather than deviations from stable person-specific trait levels.

Blimp can also fit nonlinear residual DSEM specifications. Polynomial, spline, or other functions of time can be added when the observation window or substantive process suggests nonlinear trajectories. These specifications are especially relevant in longer studies, across developmental transitions, or when repeated measurement itself may produce reactivity.

References

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